Orbital Navigation by Ultra-Advanced Civilizations: A Digression

The orbits of planets and comets in our solar system are all ellipses to very high accuracy (Figure 7.5). Newton's laws of gravity guarantee and enforce this.

By contrast, around a gigantic, spinning black hole such as Gargantua, where Einstein's relativistic laws hold sway, the orbits are far more complex. Figure 7.6 is an example. For this orbit, each trip around Gargantua would require a few hours to a few days, so the entire pattern in Figure 7.6 would be swept out in about a year. After a few years, the orbit would pass near most any destination you might wish, though the speed at which you arrive might not be right. A slingshot might be needed to change speed and make a rendezvous.

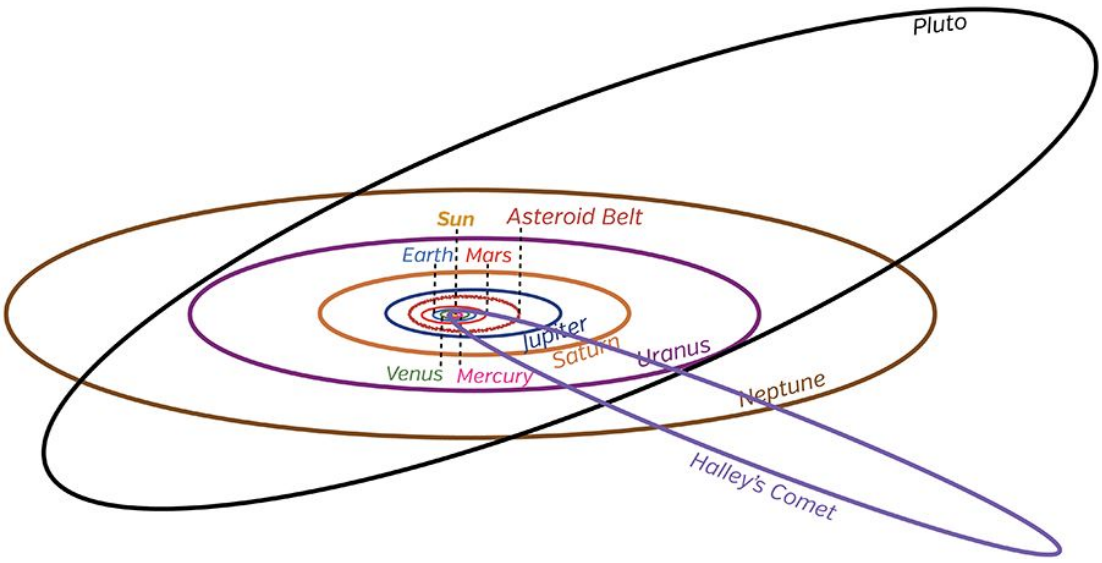


Fig. 7.5. The orbits of planets, Pluto, and Halley's comet in our solar system are all ellipses.